

Project Design Phase-II: Solution Requirements es (Functional & Non-functional)

Date	2026-09-28
Team ID	SWTID-2026-2362
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed automated laptop procurement solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Catalog Intake & Persona Filtering	• Request submission via ServiceNow Service Portal. • Automated hardware profile mapping based on User Department. • Auto-population of User Profile location and Cost Center data.
FR-2	Inventory & Stock Validation	• Synchronous database lookup against CMDB table (alm_hardware). • Branching logic validation to check local stock status. • Automated purchase order creation via IntegrationHub if stock < 10 units.
FR-3	Dynamic Approval Routings	• Direct approval request generation to employee's Line Manager. • Actionable message rendering inside MS

		Teams integration layers. • Dynamic notification routing via corporate email.
FR-4	Device Provisioning & Asset Handover	• Webhook triggers to cloud deployment platform (Windows Autopilot). • Real-time courier tracking updates pushed straight to portal logs. • Auto-transition of asset deployment status from <i>Available</i> to <i>In Use</i>.
FR-5	Lifecycle Closure & Feedback	• Asset serial identifier binding directly to active user account metadata. • Triggering micro-feedback modular survey on day seven post-delivery. • Automatic system closure parameters activation on survey submission.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution framework.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The ServiceNow Service Portal interface must allow end-users to successfully configure and submit a laptop request in under three clicks.
NFR-2	Security	All automated approval parameters and asset status data changes must restrict runtime read/write capabilities via platform Access Control Lists (ACLs).
NFR-3	Reliability	The Flow Designer engine must process transaction payloads with zero script error blocks, executing a fallback notification routing array if systems go offline.

NFR-4	Performance	The synchronous inventory lookup block running on the database layers must process asset records and complete operations within 2 seconds of catalog submit actions.
NFR-5	Availability	The automated workflow deployment environment must ensure a 99.9% platform runtime window, ensuring zero process delays during off-hours operations.
NFR-6	Scalability	The structural architecture layout must seamlessly orchestrate over 5,000+ simultaneous global laptop allocations monthly across multiple vendor pipelines.